



- INDUSTRIAL ORGANIZATION -
PRODUCTION SCHEDULING AND
OPTIMIZATION

Course goals

- Understand the definitions and the difference between centralized and decentralized scheduling
 - Know the different priority rules used for scheduling (local or global)
 - Be able to perform the algorithms used in centralized scheduling



Summary

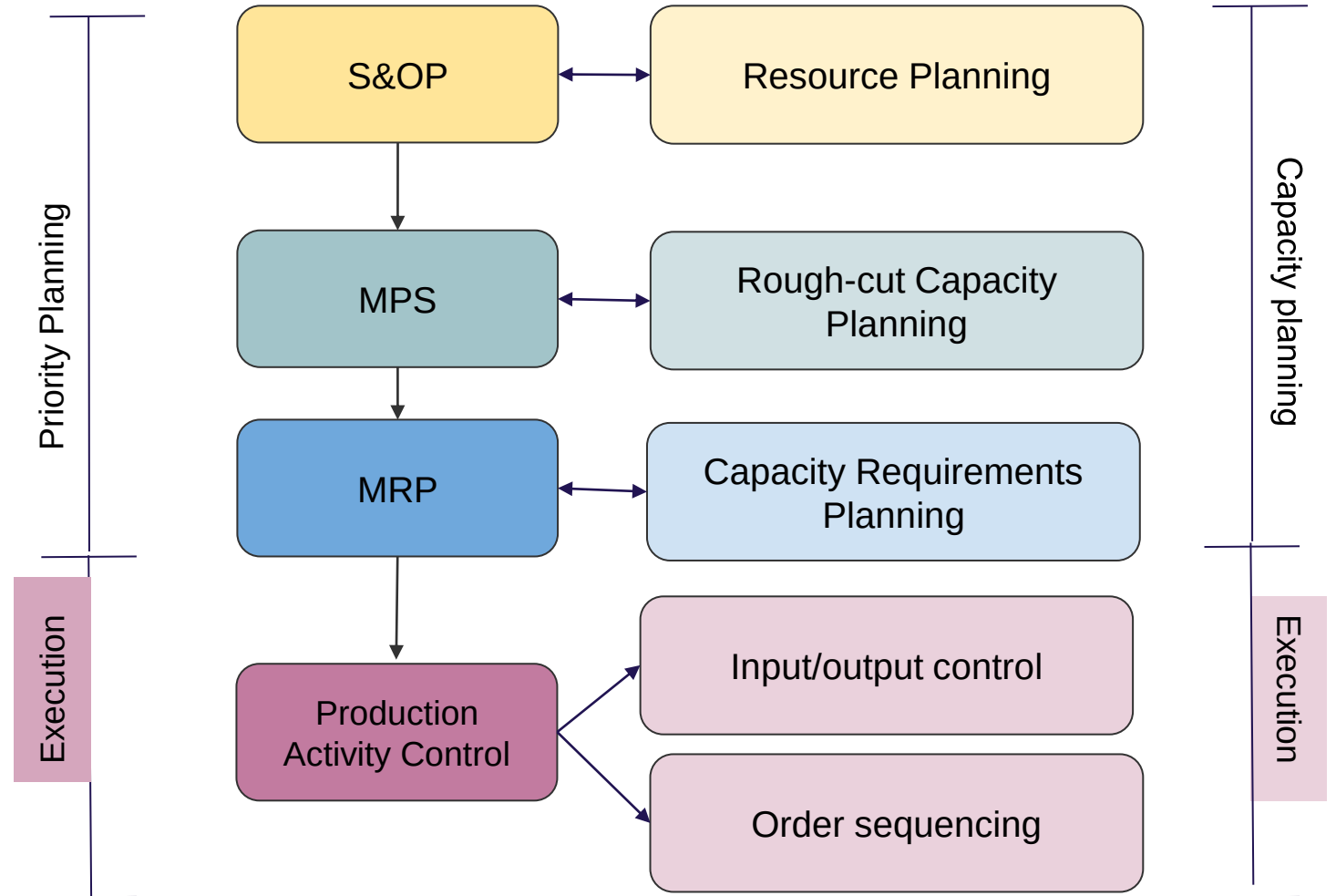
1. INTRODUCTION
2. CENTRALIZED SCHEDULING
3. DECENTRALIZED SCHEDULING

1. INTRODUCTION

Reminder of the context: Industrial planning

At each level of the industrial planning, we need to **make sure** that we will have **enough resources** to achieve the production plan.

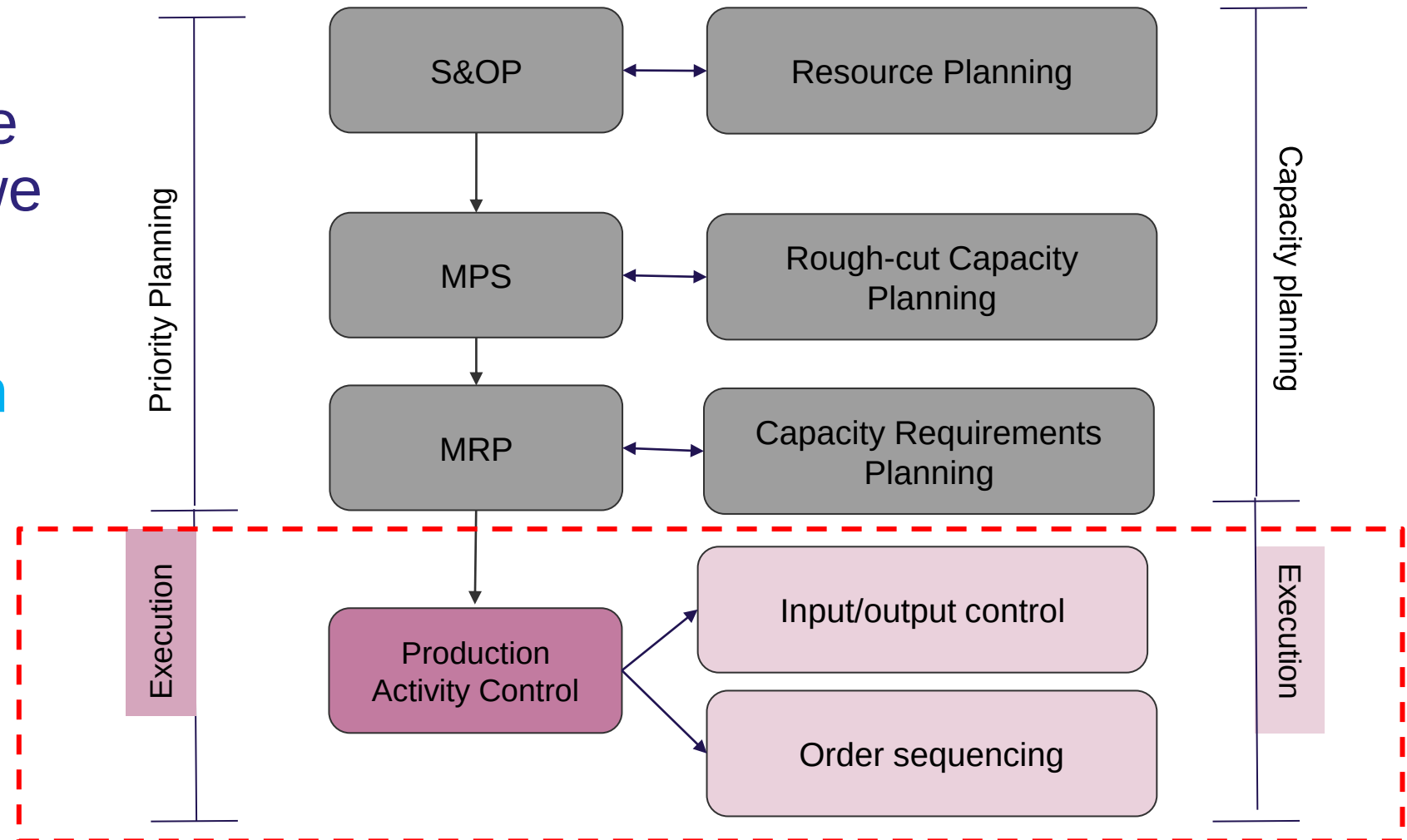
We need to check **how much capacity is available and how much is required**, before performing a **planning of the resources** based on the production schedule.



Today's topic of interest: The **execution** of the plan

Once we enter to the execution horizon, we don't speak about "planning" anymore but about **execution monitoring and follow-up**.

Our focus today will be only on this **Execution**



Priority planning v/s production scheduling: different scopes

Contrary to priority planning, production scheduling focuses on a single facility within a shorter horizon.

- ⇒ Scope of a finite-capacity production schedule:
- Within the 4 walls of a **single facility**
 - **Short- to mid-term** horizon
 - Concerned with how production order and production order operations shall be **scheduled** and **sequenced**

Priority planning v/s production scheduling: different objectives

■ Priority planning determines:

- Which items shall be produced
- When
- Where
- In which quantity

■ Production scheduling is concerned with how the production shall occur in details, on which machine, and in which specific sequence.

⇒ Production scheduling:

- Determines work order operation sequence to maximize throughput
- Considers resource and material constraints simultaneously
- Is event-based, considering continuous time, and not time buckets

Production scheduling v/s industrial planning

In a schematic way:

PLANNING

What When
(time bucket) Where How much



SCHEDULING

v/s

v/s

...



Capacity gain

Production scheduling is a very complex activity

3 typical problems that a production scheduler always has in mind:

- Need to consider a multitude of constraints:
 - Machines (capability, availability, capacity), labor (skillset, capacity, availability), material availability, changeover availability, tools, multi-stage relationship
- Always have various, and partially conflicting business objectives in mind:
 - Fill rates, changeovers, inventory, utilization, service level, ...
- Deal with unplanned event on the shop floor, reschedule every day, and update the production schedule frequently
 - Constant adaptability with a global view of the shop floor
- Be customer oriented
 - Production scheduling customers are not external but internal: usually customer service and/or supply chain departments

The importance of Data Quality and Data Accuracy

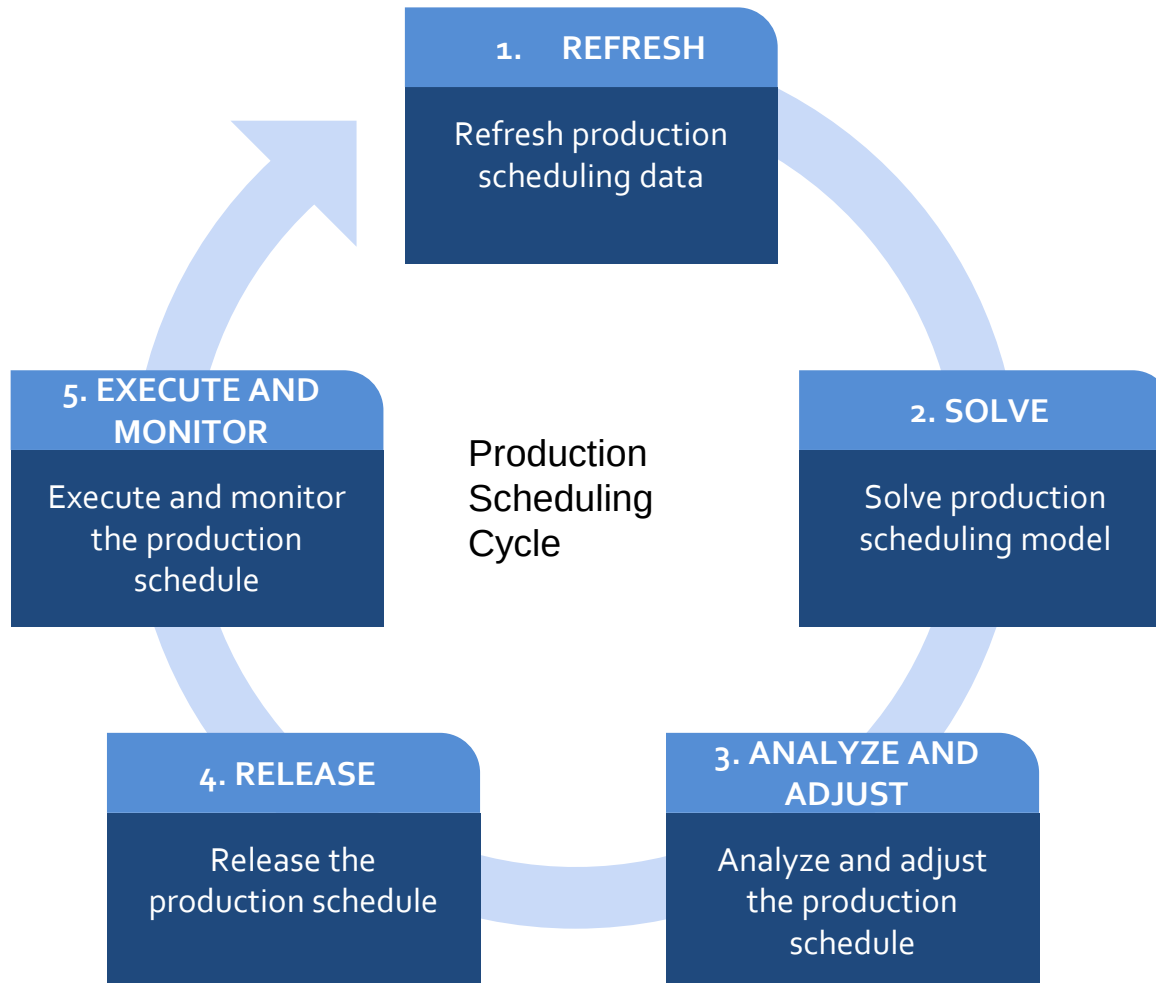
Data Quality

- Capacity and resource availability are key drivers
- **Correct routing, resource capacity, and resource availability data are essential**
- Data should be validated and updated frequently in order to be always aligned and up to date with actual execution in the shop floor

Data Accuracy

- Production order
- Tactical planning
- Some data (eg: forecasts) inherently has some degree of inaccuracy

The Production Scheduling Cycle



- Key tasks in production cycle are similar in most environments
- Cycle length may differ by industry, customer, and specific facility
- Complexity of these tasks vary, depending on the production process, the data quality and accuracy, the organization efficiency and, most importantly, the use and performance of dedicated information systems (ERP, MES, APS)

Our focus today: solving the production scheduling model

- This lecture mostly focuses on step 2 of the production scheduling cycle: solve the production scheduling model
- 2 scheduling typologies will be covered: centralized scheduling and decentralized scheduling
- For each of these typologies, scheduling techniques and rules will be explained
- The focus will be on determining the best sequence of orders in a production line in order to maximize our objectives, depending on the rules that drive the production scheduling

Definitions

The sequencing rules that we are going to define here are **not much useful for continuous or quasi-continuous production** (exemple steel production or cars production)

These rules of scheduling optimization are **mostly used is intermittent production** systems, for **small and medium production quantities** (job shop / batch production).

From the production orders calculated in MRP (if it exists in the company), or from firm or forecasted customer orders, the scheduling optimization enables to **determine the order (i.e. the sequence) of the production orders** on the different machines or work center, before launching and controlling manufacturing.

Centralized and decentralized scheduling and sequencing

2 scheduling typologies:

- Centralized scheduling & sequencing:
 - The schedule of each work center and of each machine is defined and decided **globally, at the global and central level**;
 - The distribution production order and the control of manufacturing execution are also **centralized**
- Decentralized scheduling & sequencing:
 - The decisions are taken very **locally**, by the operators themselves, based on local rules that are defined at the central level
 - The **queue of production orders** is **managed in front of each workcenter or machine**.
 - ⇒ The operators or the team manager choose the order of turn using **priority rules** that have been decided before.

2. CENTRALIZED SCHEDULING

Objectives

Centralized scheduling requires that the **routing** of the product and all the **manufacturing lead times** are precisely known and accurate.

It consists of building a **schedule (i.e. detailed planning) of the operations**, that must be aligned with the company objectives and strategy:

- Respect of the **deadline and of the commitment**
- **Service level**
- **Stock levels** (finished goods, work-in-progress, sub-assemblies, components, raw materials, ...)
- **Resources utilization optimization**

2 approaches: algorithm v/s heuristics

There are **2 approaches** to perform the production scheduling:

- Try to optimize production using **algorithms** that can help in finding the **best solution**,
- We can use a **practical approach using rules** to be able to find a **good solution** that is not necessarily the best one.

Although algorithms are powerful in many ways, and are essential in the design and development of softwares, heuristic approach is very powerful when talking about production scheduling and shop floor activities.

Those 2 approaches are different but can co-exist and are complementary.

Practical approach: Priority rules

If we don't use an algorithm approach to sequence the production orders, we have to define **priority rules that are aligned with the objectives**.

There are 5 types of priority rules, that depends on:

1. The **sequence** in which the production orders are initially positioned in the planning
2. The **lead times and routings** of the production orders
3. The **deadline** required for the production orders
4. The **cost** of production orders
5. The **bottlenecks** of the production process

Practical approach: Priority rules

Depending on the priority rules that are chosen, the **result** of the scheduling can be totally **different** (examples in the following slides).

The choice of the priority rules must **always be aligned with the production objectives**, that *should* be aligned with the company's strategy, such as:

- Customer Service Level
- Production plan adherence (respect the promised delivery date)
- Maximum load in the machines
- etc.

Practical approach: Priority rules

| Definitions

1. Rules that depend on the sequence in which the production orders are positioned in the planning
 - **First in first out (FIFO)**
 - The production orders are scheduled in their order of arrival
 - Eg.: products that have shelf life (pharmaceutical industry, food industry, etc.)
 - **Last in first out (LIFO)**
 - The last production order that arrived is scheduled first
 - Eg: screws and bolts inventory, warehouses, etc.

Practical approach: Priority rules

| Definitions

2. Rules that depend on The lead times and routings of the production orders

- Priority to the order that will go in the **shortest queue**
- Priority to the order that have the **lowest number of operations**
- Priority to the order that have the **highest number of operations**
- Priority to the order that have the **lowest manufacturing lead time**
- Priority to the order that have the **highest manufacturing lead time**

Practical approach: Priority rules

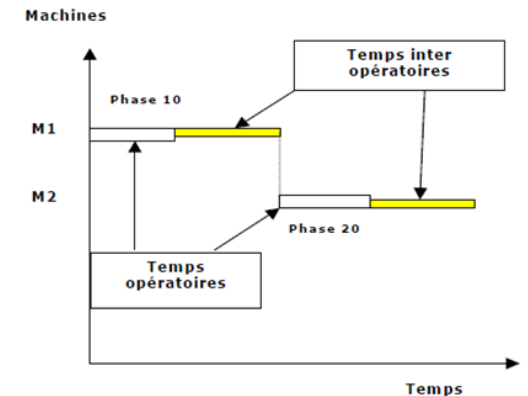
| Definitions

3. Rules that depends on The deadlines and dates required for the production orders

- **Minimum deadline**
 - We schedule first th orders that have the **earliest due date**
- **Minimum number of operation to the deadline**
- **Critical Ratio (CR):**

CR = Remaining time / Remaining work

- Remaining time = deadline - today's date
- Remaining work = Production time + inter-production time
 - Production time = Quantity x Run time + Setup
 - Inter-production time is calculated statistically (wait time between 2 operations)



Practical approach: Priority rules

| Definitions

4. Rules that depends on the cost

- Priority to the production order that **generates the highest cost**

5. Bottlenecks

- Priority to the production orders that have operations to be **performed on resources that are bottlenecks**

Examples of the different priority rules & results

Situation:

5 production orders are waiting in front of a workshop.

We are on day 60 and the we have the following **information** about the production orders:

Production Order	Day of arrival	Production time (h)	Deadline	Remaining work (d)
PO-51	53	4	66	5
PO-52	55	3	64	4
PO-53	54	5	62	3
PO-54	59	2	61	2
PO-55	60	1	65	2

Examples of the different priority rules & results

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Priority rule selected: FIFO

If we apply the FIFO rule, then the sequence of production orders will be:



Examples of the different priority rules & results

Situation:

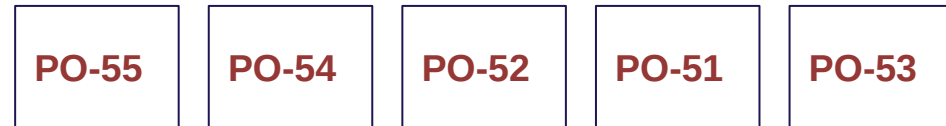
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Priority rule selected: lowest manufacturing lead time

If we apply this rule, then the sequence of production orders will be:



Examples of the different priority rules & results

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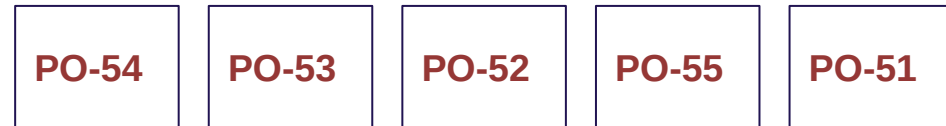
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PO-54	59	2	61	2
PO-55	60	1	65	2

Priority rule selected: Earliest due date

If we apply this rule, then the sequence of production orders will be:



Examples of the different priority rules & results

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PO-53	54	5	62	3
PO-54	59	2	61	2
PO-55	60	1	65	2

Priority rule selected: Critical ratio

Order	Deadline	Remaining time (t)	Remaining work (w)	Critical ratio (t/w)
PO-51	66	66-60=6	5	
PO-52	64	4	4	
PO-53	62	2	3	
PO-54	61	1	2	
PO-55	65	5	2	

Examples of the different priority rules & results

Situation:

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PO-54	59	2	61	2
PO-55	60	1	65	2

Priority rule selected: Critical ratio

Order	Deadline	Remaining time (t)	Remaining work (w)	Critical ratio (t/w)
PO-51	66	66-60=6	5	1.2
PO-52	64	4	4	1
PO-53	62	2	3	0.67
PO-54	61	1	2	0.5
PO-55	65	5	2	2.5

Examples of the different priority rules & results

Situation:

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PO-55	60	1	65	2

Priority rule selected: Critical ratio

Order	Deadline	Remaining time (t)	Remaining work (w)	Critical ratio (t/w)
PO-51	66	66-60=6	5	1.2
PO-52	64	4	4	1
PO-53	62	2	3	0.67
PO-54	61	1	2	0.5
PO-55	65	5	2	2.5

Order 3 and 4 will not be ended before due date

Examples of the different priority rules & results

Situation:

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PO-54	59	2	61	2
PO-55	60	1	65	2

With 4 different priority rules, you obtain 4 different results in the order scheduling.

⇒ The choice of the priority rule is very important for the management, it has to meet the company's goals.

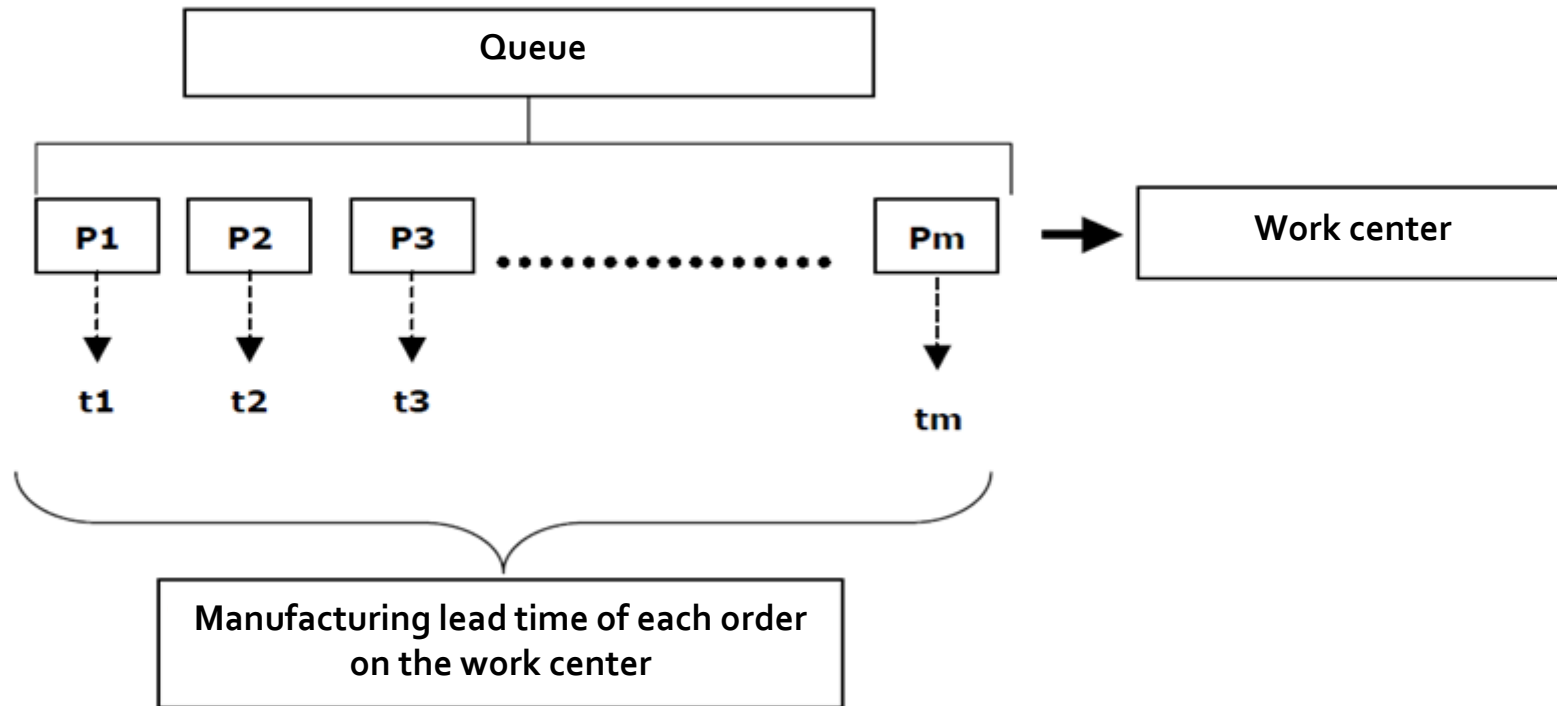
Scheduling optimization algorithm

In order to **optimize** your production schedule, one can use **algorithms** that will help you to find **the best solution** in the sequencing of your orders.

The rules that will be discussed now are global rules that aims at having the shortest **global** manufacturing lead times, taking into consideration the **full queue of production orders** - and not the production order individually.

Scheduling optimization algorithm

Shortest production lead time **globally**



Shortest average production lead time

Shortest production lead time rule enables to **determine the order of turn of the production order on the machine while minimizing the average manufacturing end date** of the production orders which are in the queue.

The lead time to the manufacturing end date of an order in the k^{th} position is the sum of all the manufacturing lead times of production orders before the k^{th} one and its own manufacturing lead time:

$$d_k = \sum_{i=1}^k t_i$$

Shortest average production lead time

If we assume all the production lead times for the different production orders : $d_1, d_2, d_3, d_4, \dots, d_m$.

The average lead time is :

$$d_m = \frac{1}{m} \sum_{j=1}^m d_j = \frac{1}{m} \sum_{j=1}^m \left(\sum_{k=1}^j t_k \right) = \frac{1}{m} \sum_{i=1}^m (m - i + 1) t_i$$

The objective is to minimize the average production lead time: d_m

⇒ In order to minimize d_m , we have to associate the shortest production lead time (t_i) to the highest coefficient ($m-i+1$), yet, this the higher i is, the higher this coefficient is.

⇒ The lowest value of d_m will be obtained if we manufacture first the orders with the shortest production lead times t_i .

Example of Shortest average production lead time

In front of a machine, we have a queue of 5 production orders with the following information :

Production Order id	Production time (h)	End of production (h)	Production deadline (h)
PO-1	8	8	16
PO-2	3	11	40
PO-3	4.5	15.5	16
PO-4	3.3	18.8	20
PO-5	5.2	24	26

Lead time to manufacture the 5
production d_5
order (k=5):

$$d_5 = \sum_{i=1}^5 t_i = t_1 + t_2 + t_3 + t_4 + t_5 = 24$$

Example of Shortest average production lead time

In front of a machine, we have a queue of 5 production orders with the following information and we want to determine the average production lead time d_m :

Production Order id	Production time (h)	End of production (h)	Production deadline (h)
PO-1	8	8	16
PO-2	3	11	40
PO-3	4.5	15.5	16
PO-4	3.3	18.8	20
PO-5	5.2	24	26
	Average lead time d_m	15.46	

$$d_m = \frac{1}{m} \sum_{i=1}^m (m - i + 1) t_i$$

$$d_m = \frac{1}{5} \times [(5 \times 8) + (4 \times 3) + (3 \times 4.5) + (2 \times 3.3) + (1 \times 5.2)] = \frac{1}{5} (77.3) = \mathbf{15.46}$$

Example of Shortest average production lead time using the sortest production lead time priority rule

If we apply the shortest production order lead time priority rule, **we have the change the order of turn**, and we obtain a different minimum average lead time of the queue :

Prod order id sorted in the Order of turn	Production time (h)	End of production (h)	Production deadline (h)
PO-2	3	3	40
PO-4	3.3	6.3	20
PO-3	4.5	10.8	16
PO-5	5.2	16	26
PO-1	8	24	16
	Average lead time time dm	12.02	

$$d_m = \frac{1}{5} \times [(5 \times 3) + (4 \times 3.3) + (3 \times 4.5) + (2 \times 5.2) + (1 \times 8)] = \frac{1}{5} (60.1) = \mathbf{12.02}$$

Example of Shortest average production lead time using the shorted production lead time priority rule

If we apply the shortest production order lead time priority rule, we have the change the order of turn, and we obtain a different minimum average lead time of the queue :

Order of turn	Production time (h)	End of production (h)	Production deadline (h)
2	3	3	40
4	3.3	6.3	20
3	4.5	10.8	16
5	5.2	16	26
1	8	24	16
	Average time	12.02	

!/ This rule doesn't take into account the deadline : order 1 is finished at t24 while is deadline is t16. !/

Johnson algorithm

Johnson algorithm aims at **minimizing the total manufacturing lead time for a queue in front of 2 machines**.

Pre-requisite: All the production orders must have the same flow (same order of turn on each machine).

While there are production orders in the queue

Find the **shortest** manufacturing lead time in the queue

If the **shortest manufacturing** lead time is on the **FIRST** machine

Then the corresponding production order must be scheduled in the first position;

Else the corresponding production order must be scheduled in the last position;

Continue until there are no more production orders in the queue

Johnson algorithm : Example

4 orders O₁, O₂, O₃ and O₄ have to be manufactured successively on machines M₁ and M₂.

We have the following routing information :

O1		
Phase	Machine	Time
10	M1	10
20	M2	8

O2		
Phase	Machine	Time
10	M1	4
20	M2	14

O3		
Phase	Machine	Time
10	M1	18
20	M2	6

O4		
Phase	Machine	Time
10	M1	20
20	M2	24

Johnson algorithm : Example

4 orders O₁, O₂, O₃ and O₄ have to be manufactured successively on machines M₁ and M₂. We have the following routing information :

O1		
Phase	Machine	Time
10	M1	10
20	M2	8

O2		
Phase	Machine	Time
10	M1	4
20	M2	14

O3		
Phase	Machine	Time
10	M1	18
20	M2	6

O4		
Phase	Machine	Time
10	M1	20
20	M2	24

Application of Johnson algorithm:

O₂ has the shortest lead time, this lead time is on M₁ ⇒ O₂ is placed in the **first position**

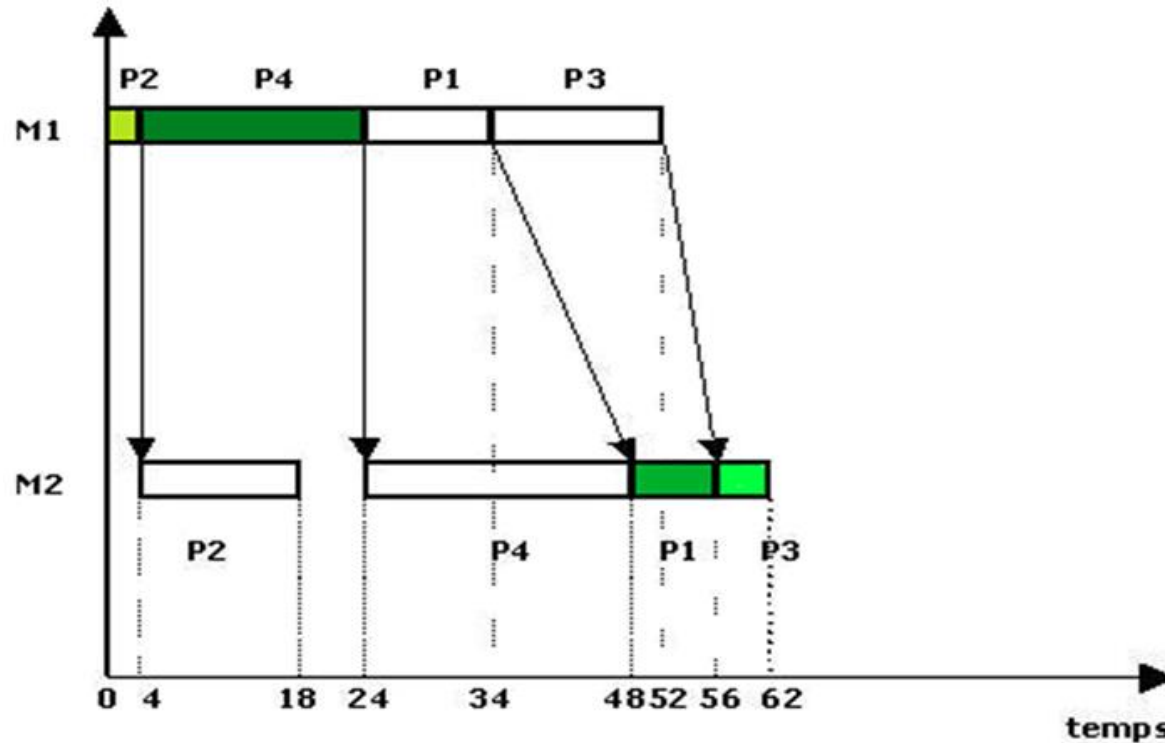
Then, within the remaining production orders, O₃ has the shortest lead time, this lead time is on M₂ ⇒ O₃ is in the **last position**

Then, within the remaining production orders, O₁ has the shortest lead time, this lead time is on M₂ ⇒ O₁ is in the **second-to-last position**

Then, only one production order remaining: O₄ ⇒ O₄ is on the **2nd position**

⇒ The result of Johnson algorithm is : **O₂, O₄, O₁, O₃**

The visual result of this algorithm is shown in a graph that represents the schedule of the queue



Johnson algorithm generalized

We can use this algorithm for **all manufacturing process** that is **sequenced on more than 2 machine/workcenter**, even if they are not all used for manufacturing.

For each production order :

- 1- Sum the lead times of all operations (N),
- 2- Sum the lead times x of the $n-1$ first operations,
- 3- Sum the lead times y of the $n-1$ last operations,
- 4- Calculate the rate $k = x/y$,
- 5- We obtain the production schedule following the increasing order of k .

Johnson algorithm generalized

Example

6 production orders are in a queue and have to be manufactured successively on 4 machines.

We have the following routing information:

P1			P2			P3			P4			P5			P6		
Phase	Machine	Time	Phase	Machine	Time	Phase	Machine	Time	Phase	Machine	Time	Phase	Machine	Time	Phase	Machine	Time
10	M1	20	10	M1	35.6	10	M1	2.8	10	M1	3.2	10	M1	24.4	10	M1	0.4
20	M2	17.2	20	M2	39.6	20	M2	18.8	20	M2	25.6	20	M2	7.6	20	M2	32
30	M3	6	30	M3	38	30	M3	8	30	M3	4.8	30	M3	26	30	M3	26.4
40	M4	1.6	40	M4	30.8	40	M4	39.2	40	M4	37.6	40	M4	5.6	40	M4	31.2

Johnson algorithm generalized

Production order	M1	M2	M3	M4	TOTAL	X	Y	K	Order of turn
P1	20	17.2	6	1.6					
P2	35.6	39.6	38	30.8					
P3	2.8	18.8	8	39.2					
P4	3.2	25.6	4.8	37.6					
P5	24.4	7.6	26	5.6					
P6	0.4	32	26.4	31.2					

Johnson algorithm generalized

1- Sum the lead times of all n operations (here n=4)

Production order	M1	M2	M3	M4	TOTAL	X	Y	K	Order of turn
P1	20	17.2	6	1.6	44.8				
P2	35.6	39.6	38	30.8	144				
P3	2.8	18.8	8	39.2	68.8				
P4	3.2	25.6	4.8	37.6	71.2				
P5	24.4	7.6	26	5.6	63.6				
P6	0.4	32	26.4	31.2	90				

Johnson algorithm generalized

2- Sum the lead times x of the $n-1$ first operations (here, $n-1 = 4 - 1 = 3$)

Production order	M1	M2	M3	M4	TOTAL	X	Y	K	Order of turn
P1	20	17.2	6	1.6	44.8	43.2			
P2	35.6	39.6	38	30.8	144	113.2			
P3	2.8	18.8	8	39.2	68.8	29.6			
P4	3.2	25.6	4.8	37.6	71.2	33.6			
P5	24.4	7.6	26	5.6	63.6	58			
P6	0.4	32	26.4	31.2	90	58.8			

Johnson algorithm generalized

3- Sum the lead times y of the $n-1$ last operations

Production order	M1	M2	M3	M4	TOTAL	X	Y	K	Order of turn
P1	20	17.2	6	1.6	44.8	43.2	24.8		
P2	35.6	39.6	38	30.8	144	113.2	108.4		
P3	2.8	18.8	8	39.2	68.8	29.6	66		
P4	3.2	25.6	4.8	37.6	71.2	33.6	68		
P5	24.4	7.6	26	5.6	63.6	58	39.2		
P6	0.4	32	26.4	31.2	90	58.8	89.6		

Johnson algorithm generalized

4- Calculate the rate $k = x/y$

Production order	M1	M2	M3	M4	TOTAL	X	Y	K	Order of turn
P1	20	17.2	6	1.6	44.8	43.2	24.8	1.74	
P2	35.6	39.6	38	30.8	144	113.2	108.4	1.04	
P3	2.8	18.8	8	39.2	68.8	29.6	66	0.45	
P4	3.2	25.6	4.8	37.6	71.2	33.6	68	0.49	
P5	24.4	7.6	26	5.6	63.6	58	39.2	1.48	
P6	0.4	32	26.4	31.2	90	58.8	89.6	0.66	

Johnson algorithm generalized

5- We **obtain the order of turn, therefore the production schedule** following the increasing order of k.

Production order	M1	M2	M3	M4	TOTAL	X	Y	K	Order of turn
P1	20	17.2	6	1.6	44.8	43.2	24.8	1.74	6
P2	35.6	39.6	38	30.8	144	113.2	108.4	1.04	4
P3	2.8	18.8	8	39.2	68.8	29.6	66	0.45	1
P4	3.2	25.6	4.8	37.6	71.2	33.6	68	0.49	2
P5	24.4	7.6	26	5.6	63.6	58	39.2	1.48	5
P6	0.4	32	26.4	31.2	90	58.8	89.6	0.66	3

Conclusion

Centralized scheduling permits to **build a schedule** for **the use of the resources** and to **provide information** about delivery dates.

However, in a workshop organization, it **can be unrealistic** because **we try to manage something uncertain and dynamic with fixed rules**.

That is why some companies prefer decentralized scheduling.

It mainly depends on:

- Organization
- Production process

3. DECENTRALIZED SCHEDULING

Objectives

Using decentralized scheduling, **we can't make any forecasts** or planning because the sequencing decisions are made in front of the machines $\Rightarrow$ therefore, it is difficult to **commit** on a delivery date to internal and external customers

When a machine has finished with a batch, **the operator can choose the next one** based on **priority rules** that has been given to him before.

The idea is : if we **apply the rules locally**, we should be able to **reach the global objectives** of the workshop.

We can use 2 type of rules :

- **Local rules** that only take into account the order of the queue in front of the machine,
- **Global rules** that take into account the other machines of the workshop.

Local rules

First in first out (FIFO)

The simplest rule, but it may not be adapted if the times between today and the deadlines are short regarding operation lead times.

Priority to the batch with the shortest manufacturing lead time

There is a risk that production orders with a long production lead time stay in the queue for a long time

Earliest deadline

It prioritizes the production orders that have a close delivery date or those that are late.

Critical ratio

The objective is to respect the deadline.

Global rules

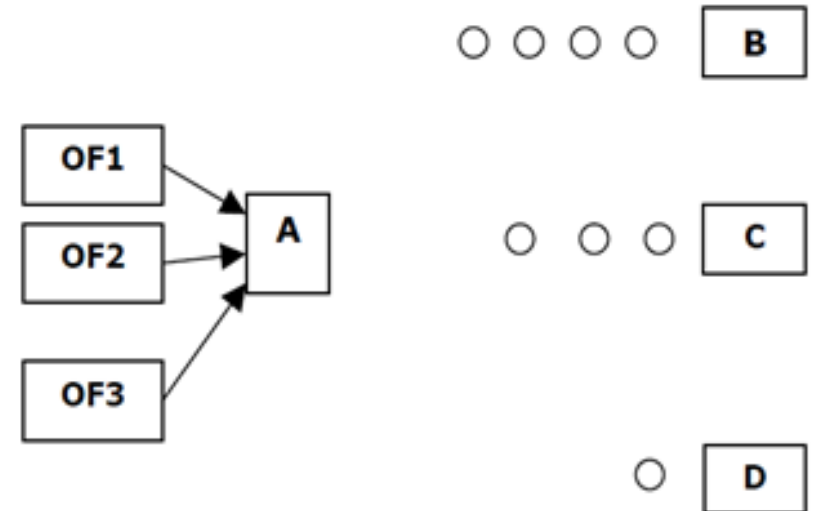
- Information about all the queues in the workshop are necessary to apply these rules.
- Thus, these information has to be centralized, even if the decision is made locally.

Global rules

- Priority to the production order that will go in the shortest queue for the next operation

In the example, we will chose OF₃ first because the next step for it is D, which has the shortest queue, it permits to avoid a stop in production on D and to add an order in the other queues.

	OF1	OF2	OF3
PH 10	A	A	A
PH 20	B	C	D



- Priority to orders which next step is on a bottleneck

Thank you for your attention



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